| Team Number : | apmcm24203506 |
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| Problem Chosen : | C |

2024 APMCM summary sheet

# Pet market analysis model based on multiple regression and grey prediction models

## Summary

**Pet industry is a diversified industrial chain formed around pet raising, consumption and service. The pet market covers a wide range of economic industries, including live pet trading, pet supplies, pet food, pet medical treatment, pet training and so on. Considering economic, social, scientific research and other factors and data, this paper establishes a regression model to predict the relevant situation of China's and the global pet industry, especially the regression model of the development trend of the pet food industry, which accounts for more than half of the industry, and actively puts forward suggestions for the sustainable development of the industry.**

**In** response to question 1**, the development trend of China's pet industry from 2019 to 2023 was analyzed** through normality test**. According to the changes in the number of pet cats and dogs in China and the penetration rate of pet cats and dogs in households, the Pearson correlation coefficient was used to analyze the relevant influencing factors, and the partial least square method + gray prediction mathematical model was established to predict the development of the industry in the next three years. The number of pet cats is expected to reach 147.71 million in 2026; Dog populations are more stable, with small increases or decreases, and are expected to reach 50.69 million in 2026. Factors influencing the growth include urbanization rate, per capita disposable income growth, and the number of pet policies at the national level.**

- **In response to question 2, considering the development of pet industry in different countries, we first find four representative countries, the United States, Germany, France and China, which are among the top countries in the world's pet market, and analyze the development of global pet industry according to different pet types; Then, based on Spearman correlation coefficient analysis, we screened the economic level, pet food industry development data and other quantitative data of pet industry development of major countries. Considering the accuracy and robustness of the model, the pet food market size of each country represented the industry demand. The 2024-2026 pet food demand prediction models for different countries in 2019-2023 were established by SVR regression + gray prediction algorithm. Python language was used for data pre-processing, and SPSS was used to complete the regression model establishment and result analysis. Then, the proportion of major countries in the global pet food market from 2019 to 2026 was predicted by gray scale, and the scale trend of global pet food was obtained by removing the corresponding proportion.**
- **In view of the third question, first of all, combined with the production and export volume of China's pet food in the past five years, the development trend of China's pet food industry is analyzed. Then, the relationship between the global pet food market demand and China's pet food production and export is discussed. Firstly, the scale of global pet food market is selected as the quantitative index of market demand. At the same time, from the aspects of China's economic development, social development, scientific and technological development, market development trend and transportation situation, The per capita disposable income of China's total import and export of goods in the past five years, the transaction scale of online retail industry and the number of valid patent grants in China's**

**pet industry were selected as the indicators to quantify China's development. Spearman correlation analysis was used to select the quantitative indicators highly correlated with China's pet food production and export volume as the independent variables of the regression model. In view of the small sample size and complex relationship between features and dependent variables, SVR (support vector regression) and PLS (partial least squares regression) models were used to construct multiple regression models, and the regression fitting equations of production volume and export volume were obtained respectively. In terms of time dimension, the quantified index values from 2024 to 2026 were predicted by the gray prediction model, and the production and export of pet food in China in the next three years were predicted by using z_score standardized processing into the regression equation. The main modeling tools are Python and SPSS.**

- **In response to question 4, China's pet food market is not only affected by the economic and social factors mentioned in the previous questions, but also by the relevant economic policies of overseas countries such as Europe and America. We can establish a mathematical model according to various data we can find, and quantitatively analyze the impact of these economic policies on China's pet food market. Based on the analysis results, we can further find out the feasible strategies to cope with the changes of foreign economic policies, so as to achieve the long-term development of China's pet food industry.**

**Keywords: Pet market; Partial least squares; Grey prediction; Regression fitting**

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# 1 Introduction

## 1.1 Problem Background

**China's pet industry is a diversified industrial chain formed around pet raising, consumption and service. The pet market covers a wide range of economic industries including live pet trading, pet supplies, pet food, pet medical treatment, pet training and so on, involving a wide range of industrial chains. The development of China's pet industry is related to the emotional needs of the vast number of Chinese people, and sustenance the people's growing need for a better life. At the same time, there are many factors affecting the development trend of the pet industry, which is closely related to the development situation of science and technology, society, economy, communication and other aspects. The research on the development of the pet industry can reflect the process of Chinese-style modernization in many fields. Analyzing and forecasting the development of China's pet industry and the global pet industry, especially the pet food industry, which accounts for more than half of the industry, is conducive to a deeper understanding of China's development and international influence from multiple perspectives.**

## 1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- Problem 1, based on the data provided in Annex 1 and the additional supplementary information collected, a detailed analysis of the development situation by pet type in China in the past five years is conducted. At the same time, explore the key factors affecting the development of China's pet industry in order to build a suitable mathematical model to predict the development trend of China's pet industry in the next three years.
- Problem 2,
- Problem 3 is to analyze the development of China's pet food industry by observing the production and export data trends of China's pet food in the past five years, and taking into account the impact of the increase or decrease in global pet food market demand and the development of China's relevant fields on the pet food industry without considering policy factors. Predict the trend of China's pet food production and export value in the next three years.
- Problem 4: In addition to the economic and social factors mentioned in the previous questions, China's pet food market is also affected by the relevant economic policies of overseas countries such as Europe and America. We can build a mathematical model based on all kinds of data we can find, and quantitatively analyze the impact of these economic policies on China's pet food market. Based on the analysis results, we can further find out the feasible strategies to cope with the changes of foreign economic policies, so as to achieve the long-term development of China's pet food industry.

## 1.3 Our Work

### 1.3.1 Analysis of Problem One

In this paper, the number of pet cats and dogs in China and household penetration rate are used to describe the development of China's pet industry. Pearson correlation coefficient is used to conduct correlation test on the data to determine whether linear model or non-linear model is used and whether there are multiple linear relationships among influencing factors. The data is tested by the level ratio to determine whether it is suitable for the gray prediction model. The grey prediction model is established, and the values of the influencing factors in the next three years are predicted through the data from 2019 to 2023. The partial least squares model was established to analyze the correlation between the dependent variable and the influencing factor. The number of pet cats and dogs and the household penetration rate in China in the next three years were predicted by the correlation and the predicted value of the influencing factor in the next three years. The data was used to analyze the development of China's pet industry.

### 1.3.2 Problem two analysis

In order to forecast the demand of global pet food market, we have studied the development of pet industry in different countries in depth. First of all, the United States, Germany, France and China, which are the leading countries in the pet market, are selected as representatives, and the raising situation of various pets (such as cats, dogs, etc.) and the consumption trend of pet food in these countries are analyzed in detail. On this basis, Spearman correlation coefficient analysis was used to screen out characteristic variables highly correlated with pet food market size from economic level, pet food industry development data and other quantitative data of pet food industry. In order to improve the accuracy and robustness of the prediction model, SVR regression algorithm was selected and combined with grey prediction algorithm to establish a 2024-2026 pet food demand prediction model based on the data of various countries from 2019-2023. Finally, the share of major countries in the global pet food market is predicted, so as to obtain the trend of the global pet food market size.

### 1.3.3 Question 3 Analysis

The third question can be solved in two small parts. The first part is to analyze the development of China's pet food industry by combining the production and export volume of China's pet food in the past five years. According to the close relationship between supply and demand, the change trend of the overall market demand of China's pet food industry can be obtained by analyzing the change of production amount, namely the change trend of supply side production, so as to reflect the development of productivity and market demand of China's pet food industry. The change trend of export volume can represent the development trend of China's pet food industry in overseas markets, so as to analyze the overseas influence and competitiveness of China's pet food industry.

In the second part, considering the demand of the global pet food market and the development of China, the regression model between the production and export volume of pet food in China is established by selecting suitable quantitative indicators. Firstly, the global pet food market size in five years is selected to approximate the global pet food market demand. In the selection of quantitative indicators of China's development, we consider economic development, social development, scientific and technological development, market development trend, transportation and so on. Per capita disposable income was selected to quantify economic development, urbanization rate to quantify social development, total import and export volume of goods in China in the past five years to quantify transportation strength, transaction scale of online retail industry to quantify market development in the Internet era, and the number of valid patent authorization in China's pet industry to quantify the impact of scientific and technological strength development in the pet market. Then Spearman correlation analysis was used to select quantitative indicators that were highly correlated with China's pet food production and export volume as independent variables for regression. In terms of regression model selection, considering the small sample size in the past five years and the difficulty in determining the complex f relationship between multiple features and dependent variables, SVR(support vector regression) and PLS (partial least squares regression) models were selected to construct multiple regression models between the features of each quantitative indicator and the amount

of production and export value, and the regression fitting equation was obtained. In the dimension of time change, the predicted value of each quantitative index from 2024 to 2026 was obtained through the gray prediction model, and the production and export situation of China's pet food in the next three years was obtained by substituting the predicted value of the characteristics of the quantitative index into the regression fitting equation.

### 1.3.4 Problem 4 Analysis

In order to quantitatively analyze the impact of economic policies on China's pet food market, we use the exchange rate of USD to RMB and other relevant factors to construct a partial least squares model to predict the export volume of pet food in China, and then keep other factors unchanged and only change the exchange rate, so as to obtain the quantitative impact of changes in the exchange rate of USD to RMB on the export volume of pet food in China. And it is approximately regarded as the impact on the pet food market in China.

## 2 Assumptions and Justifications

1. Assume that all kinds of data found are true data of the current year, and there are no false and predicted data.
2. When establishing the regression equation, consider the overall trend, and exclude the influence of some special errors and fluctuating data on the overall model construction.
3. Only the core factors in the problem are considered. For example, when analyzing the influence of policy on the market size trend, other factors that affect the market size are not considered at the same time, and the influence of minor factors is excluded.

## 3 Notations

The key mathematical notations used in this paper are listed in Table 1.

**Table 1: Notations used in this paper**

| Symbol | Description | Unit |
|---|---|---|
| $C_1^i$ | For the number of pet cats in China, i=0 is the original data and i=1 is the generated data | |
| $D_1^i$ | Number of pet dogs in China | |
| $C_2^i$ | The penetration rate of pet cats in Chinese households | |
| $D_2^i$ | Household penetration of pet dogs in China | |

# 4 The name of model 1

## 4.1 The Establishment of Model 1

### 4.1.1 Construct the grey prediction model

GM(1,1) is a prediction model suitable for small sample, weak information background, using the level ratio test Annex I and additional lookup data, the result can be used GM(1,1) model. By using the grey prediction model, the data for the next three years for influencing factors such as per capita disposable income and urbanization rate are predicted.

By Accumulated Generating Operation (AGO), original data series can be converted into generation series with certain regularity.

Original data sequence: ={},$M_j^{(0)}M_j^0(1)$、$M_j^0(2)$，$M_j^0(3)$，$M_j^0(4)$，$M_j^0(5)$

M=C for cat; M=D for dog; j=1 number of tables; j=2 table household penetration,; 1 to 5 Table 2019 to 2023

Generating sequence: ={}$M_j^{(1)}M_j^1(1)$、$M_j^1(2)$，$M_j^1(3)$，$M_j^1(4)$，$M_j^1(5)$

Where,$M_j^1(1) = M_j^1(1)$

$$M_j^1(2) = M_j^1(1) + M_j^1(2)$$

…

$$M_j^1(5)=M_j^1(1) + M_j^1(2) + \cdots + M_j^1(5)$$

Define the grey derivative as $M_j^{(1)}$

$$d(k) = M_j^{(0)}(k) = M_j^0(k) - M_j^0(k-1)，\ \mathrm{k} = 2,3,4,5 \tag{1}$$

make$Z_j^1(k) = aM_j^0(k) + (1-a)M_j^0(k-1)$

Construct a differential equation where the data satisfies a first-order linear differential equation:

$$\frac{dM_j^{(1)}}{dt} + M_j^{(1)} = b \tag{2}$$

After discretization:

$$M_j^{(0)} + aM_j^1(k) = b \quad ,k = 2,3,4,5 \tag{3}$$

a is the development coefficient and b is the grey action

Model test:

Posterior difference ratio test: Posterior difference over C value, which is the ratio of the variance of the residual to the variance of the data. The smaller the C value, the higher the

accuracy of the fit of the model.

Model residual test: relative error value and stage ratio deviation value. The smaller the relative error value, the better. Generally, less than 20% means that the fit is good.

**4.1.2 Partial least squares model construction**

Due to the multiple linear relationships among the influencing factors, a partial least squares model was constructed to find the relationship between the development situation and the influencing factors by using the data from 2019 to 2023, and to predict the number of pet cats and dogs and household penetration rate in the next three years

(1) The data was standardized for n times to obtain a standardized matrix

$$F_0 = \begin{bmatrix} y_{1C_1} & y_{1C_2} & y_{1D_1} & y_{1D_2} \\ y_{2C_1} & y_{2C_2} & y_{2D_1} & y_{2D_2} \\ \cdots & \cdots & \cdots & \cdots \\ y_{nC_1} & y_{nC_2} & y_{nD_1} & y_{nD_2} \end{bmatrix}, E_0 = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,10} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,10} \\ \cdots & \cdots & \cdots & \cdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,10} \end{bmatrix}$$

Where, denotes the number of pet cats in China, denotes the number of pet cats in China, denotes the household penetration rate of pet cats in China, denotes the household penetration rate of pet dogs in China, and denotes the i th variable of the NTH standardization.$C_1 D_1 C_2 D_2 x_{n,i}$

(2) Extract the first pair of components of the two-variable group

Suppose that the first pair of components from the two groups of variables are respectively sum, which is a linear combination of the dependent variable set:, is a linear combination of the independent variable set: =. $u_1 v_1 u_1 Y = \left(y_{1C_1}, y_{1C_2}, y_{1D_1}, y_{1D_2}\right)^T u_1 = w_{11} y_{1C_1} + w_{12} y_{1C_2} + w_{13} y_{1D_1} + w_{14} y_{1D_2} = w_1^T Y z_1 X = \left(x_{1,1}, x_{1,2}, \ldots, x_{1,10}\right)^T z_1 = v_{11} x_{1,1} + v_{11} x_{1,2} + \cdots + y_{1D_2} = v_{1/10} x_{1,10} v_1^T X$For the purpose of regression analysis, satisfy:

$$\langle \hat{t}_1, \hat{u}_1 \rangle = \langle E_0 v_1, F_0 w_1 \rangle = v_1^T E_0^T F_0 x_1 \Rightarrow max$$
$$w_1^T w_1 = \|w_1\|^2 = 1, v_1^T v = \|v_1\|^2 = 1$$

(3) Establish the correct regression and the correct regression.$y_{1C_1}, y_{1C_2}, y_{1D_1}, y_{1D_2} z_1 x_{1,1}, x_{1,2}, \ldots, x_{1,10} z_1$

$$\begin{cases} E_0 = \hat{z}_1 \hat{z}_1 + \|\hat{z}_1\|^2{}_1 \\ F_0 = \hat{u}_1 \hat{z}_1 + \|\hat{z}_1\|^2 \end{cases}$$

Where,, is the regression coefficient vector, and is the residual matrix, the least square estimation of the regression coefficient vector is: $\alpha_1 = \left(\alpha_{1,1}, \alpha_{1,2}, \ldots, \alpha_{1,10}\right)^T \beta_1 = (\beta_{11}, \beta_{12}, \beta_{13}, \beta_{14})^T E_1 F_1$

$$\begin{cases} \alpha_1 = \hat{z}_1 E_0^T + E_1 \\ \beta_1 = \hat{z}_1 F_0^T + F_1 \end{cases}$$

$\alpha_1$, is the model effect load$\beta_1$

If, approaching 0, the model accuracy is satisfied, stop extracting components. $E_1 F_1$Otherwise, use, instead of the above, and extract components for the second group of variables.$E_1 F_1 E_0 F_0$

There are r components in (5),, such that$E_0 z_1, z_2, \ldots, z_r$

$$\begin{cases} E_0 = \hat{z}_1\hat{\alpha}_1^T + \cdots + \hat{z}_r\hat{\alpha}_r^T + E_r \\ F_0 = \hat{z}_1\hat{\beta}_1^T + \cdots + \hat{z}_r\hat{\beta}_r^T + F_r \end{cases}$$

Plug in and get the partial least squares regression equation: $z_k = v_{k1}x_{1,1} + \cdots + v_{k10}x_{1,10}(k = 1,2,\ldots,r), Y = z_1\,\beta_1 + \cdots + z_r\,\beta_r$

$$y_j = a_{j1}x_{1,1} + \cdots + v_{j10}x_{1,10}, (j = 1,2,\ldots,10)$$

Model check:

Cross validity test is adopted: the i th observation is dropped each time (i=1,2,... ,n), the remaining n-1 observations are modeled according to partial least squares regression method, and the regression formula fitted after extracting h components is considered, and then the i th observation point omitted is substituted into the fitted regression equation to obtain (j=1,2,... $y_j$,p) predicted value at the i th observation point. For i=1,2,... ,n repeat the above verification and observe the sum of squares of the prediction error, when its value is the lowest corresponding h is the number of components.

4.2.3 Improved model of z-score standardization

Since the partial least squares model adopts the result after the standardization of the original data z-score, this problem needs to standardize the predicted value obtained by the gray prediction and the data used in the construction of the model.

For each item of raw data, z-score standardization formula is as follows:

$$z = \frac{x - mean}{std} \tag{3}$$

Where z is the normalized value of the Z-score of the raw data, x is the raw data, mean is the mean of the raw data, and std is the standard deviation of the raw data.

Therefore, in order to obtain the z-score standardized value of the forecast data, it is necessary to first find the mean value and standard deviation of each original data, and then substitute the forecast data into the improved model of z-score standardization. The z-score standardized value defined as the improved forecast data is the forecast data. For each forecast data, the formula of the improved z-score standardized value is as follows: $z_p\ x_p$

$$z_p = \frac{x_p - mean}{std} \tag{3}$$

## 4.2 The Solution of Model 1

### 4.2.1 Grey prediction model solution

Through the least square method, the matrix equation is constructed:

$$B = \begin{bmatrix} -M_j^1(2) & 1 \\ -M_j^1(3) & 1 \\ -M_j^1(4) & 1 \\ -M_j^1(4) & 1 \end{bmatrix}, Y = \begin{bmatrix} M_j^0(2) \\ M_j^0(3) \\ M_j^0(4) \\ M_j^0(5) \end{bmatrix} \tag{4}$$

By solving, get parameters a,b$[a,b]^T = (B^TB)^{-1}B^TY$

The parameters a= -0.125 and b= 764.046 are obtained for the data of consumer market size over the years

The parameter a= -0.0044 is obtained for the cumulative number of current national development policies and planning of the pet product industry.

b= 59.791. The same is true for other influencing factor parameter values.

The prediction formula is obtained as follows:

$$\widehat{M}_j^{(1)}(k+1) = \left(M_j^1(1) - \frac{b}{a}\right)e^{-ak} + \frac{b}{a} \qquad ,k = 0,1,2,3,4 \quad (5)$$

The generated sequence is restored to obtain the predicted value of the original sequence:

$$\widehat{M}_j^{(0)}(k+1) = \widehat{M}_j^{(1)}(k+1) - \widehat{M}_j^{(1)}(k) \quad (6)$$

For example, the market size of pet dogs from 2024 to 2026 is 1642.568, 1756.348 and 1878.01, as detailed in the attachment.

**4.2.2 The partial least squares model is solved**

The factor load coefficient is obtained by substituting the data

Table 1: Factor load coefficient of pet cat data

| Variables | Factor 1 | Factor 2 |
|---|---|---|
| Urbanization Rate | 0.236 | 0.794 |
| Proportion of the Single Population Aged 15 and Above in the Total Population | 0.331 | 0.18 |
| Cumulative Number of Current National-level Development Policies and Plans for the Pet Industry | 0.329 | -0.004 |
| Per Capita Disposable Income of Residents | 0.33 | -0.204 |
| Proportion of the Population Aged 65 and Above | 0.335 | -0.08 |
| Transaction Scale of the Online Retail Industry | 0.335 | -0.065 |
| Number of Valid Patent Authorizations in the Pet Industry | 0.329 | -0.162 |
| Number of Financing Events in the Pet Industry | -0.302 | -0.468 |
| Number of Newly Established Enterprises with Business Scope Containing "Pet" | 0.316 | -0.213 |
| Number of Licensed Veterinarians | 0.335 | -0.04 |
| Number of cat | 0.227 | -0.784 |
| Scale of the Consumption Market of Cat | 0.329 | -0.189 |

Table 2: Factor load coefficients for pet dog data

| Variables | Factor 1 | Factor 2 |
|---|---|---|
| Urbanization Rate | 0.243 | 0.768 |
| Proportion of the Single Population Aged 15 and Above in the Total Population | 0.33 | 0.137 |
| Cumulative Number of Current National-level Development Policies and Plans for the Pet Industry | 0.326 | -0.047 |
| Per Capita Disposable Income of Residents | 0.325 | -0.246 |
| Proportion of the Population Aged 65 and Above | 0.331 | -0.121 |
| Transaction Scale of the Online Retail Industry | 0.331 | -0.11 |
| Number of Valid Patent Authorizations in the Pet Industry | 0.324 | -0.202 |
| Number of Financing Events in the Pet Industry | -0.304 | -0.436 |
| Number of Newly Established Enterprises with Business Scope Containing "Pet" | 0.31 | -0.26 |
| Number of Licensed Veterinarians | 0.332 | -0.08 |
| Number of dogs | -0.258 | -0.561 |
| Scale of the Consumption Market of Dog | 0.268 | -0.649 |

### 4.2.3 Improvement model solution for z-score standardization:

Table 3: Improved results of z-score standardization of pet cat prediction data

| Years | 2024 | 2025 | 2026 |
|---|---|---|---|
| Urbanization Rate | 0.318899138 | 0.412008375 | 0.59822685 |
| Proportion of the Single Population Aged 15 and Above in the Total Population | 1.892459238 | 2.60124547 | 3.451788948 |
| Cumulative Number of Current National-level Development Policies and Plans for the Pet Industry | 2.043015674 | 2.809146552 | 3.830654388 |
| Per Capita Disposable Income of Residents | 2.039500875 | 2.965765067 | 3.952701159 |
| Proportion of the Population Aged 65 and Above | 2.051520866 | 2.896264752 | 3.861686336 |
| Transaction Scale of the Online Retail Industry | 2.076631873 | 3.056398451 | 4.120586625 |
| Number of Valid Patent Authorizations in the Pet Industry | 2.009920741 | 2.977750924 | 4.041849322 |
| Number of Financing Events in the Pet Industry | -1.27564081 | -1.693883698 | -2.112126586 |
| Number of Newly Established Enterprises with Business Scope Containing "Pet" | 2.139240213 | 3.302707447 | 4.662329116 |
| Number of Licensed Veterinarians | 2.060094017 | 3.006382233 | 4.021166528 |

Table 4: Improved results of z-score standardization of pet dog prediction data

| Years | 2024 | 2025 | 2026 |
|---|---|---|---|
| Urbanization Rate | 0.318899138 | 0.412008375 | 0.59822685 |
| Proportion of the Single Population Aged 15 and Above in the Total Population | 1.892459238 | 2.60124547 | 3.451788948 |
| Cumulative Number of Current National-level Development Policies and Plans for the Pet Industry | 2.043015674 | 2.809146552 | 3.830654388 |
| Per Capita Disposable Income of Residents | 2.039500875 | 2.965765067 | 3.952701159 |
| Proportion of the Population Aged 65 and Above | 2.051520866 | 2.896264752 | 3.861686336 |
| Transaction Scale of the Online Retail Industry | 2.076631873 | 3.056398451 | 4.120586625 |
| Number of Valid Patent Authorizations in the Pet Industry | 2.009920741 | 2.977750924 | 4.041849322 |
| Number of Financing Events in the Pet Industry | -1.27564081 | -1.693883698 | -2.112126586 |
| Number of Newly Established Enterprises with Business Scope Containing "Pet" | 2.139240213 | 3.302707447 | 4.662329116 |
| Number of Licensed Veterinarians | 2.060094017 | 3.006382233 | 4.021166528 |

### 4.2.4 The final result is solved:

By substituting the improved results of z-score standardization of the predicted data into the results of partial least squares model solving, the development situation of China's pet market in the next three years can be obtained as follows:

Table 5: The development of China's pet market in the next three years

| Years | 2024 | 2025 | 2026 |
|---|---|---|---|
| Number of Dogs | 5157 | 5118 | 5069 |
| Number of Cats | 9805 | 12187 | 14771 |
| Scale of the Dog Consumption Market | 1698.081354 | 1862.900022 | 2045.547118 |
| Scale of the Cat Consumption Market | 1533.344383 | 1754.021952 | 2001.606572 |

Compare the above results with those obtained by the grey prediction model as follows:

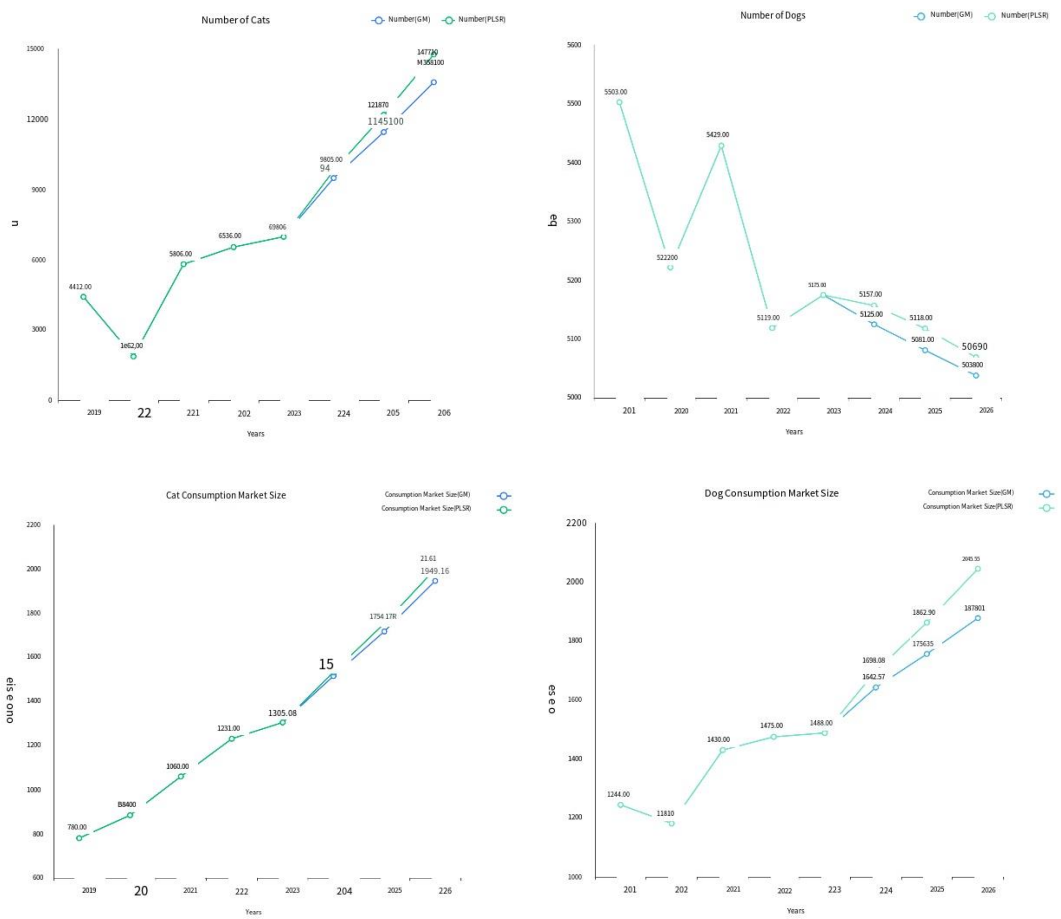



**Figure 41 Comparison of two prediction models4-1**

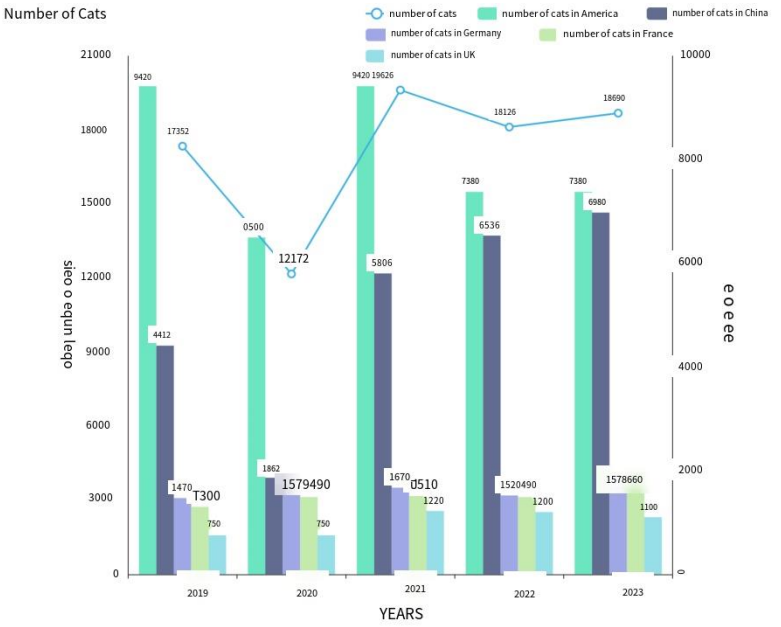
Table 11

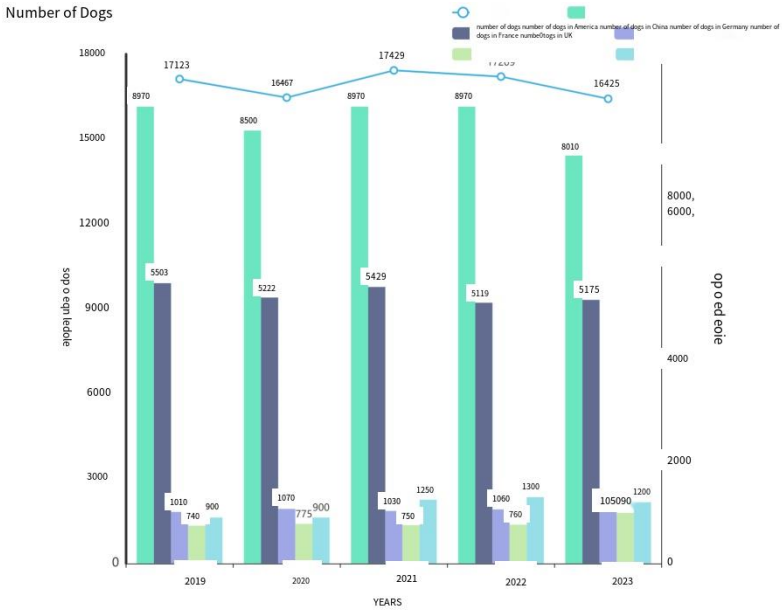
| | R2(PLSR) | Average relative error (GM) |
|---|---|---|
| Number of Cats | 0.947 | 20.85% |
| Scale of the Consumption Market of Cat | 0.98 | 2.463% |
| Number of Dogs | 0.849 | 1.333% |
| Scale of the Consumption Market of Dog | 0.976 | 3.655% |

The closer the R-$^2$value is to 1, the better, and a general value greater than 0.7 is generally considered a relatively good model. The smaller the relative error value, the better, and generally less than 20% indicates a good fit. From the above data, it can be seen that the prediction effect of partial least squares model is generally better than that of gray prediction model.

## 4.3 Question 2

In order to analyze the development of the global pet industry by pet type, we collected the population changes of two major pets in the world -- cats and dogs -- in some countries from 2019 to 2023, and drew the following line chart for visual analysis.




Cats: As can be seen from the above line chart, in Germany, France and the United Kingdom, the number of cats fluctuates, but the overall trend is on the rise; While the number of cats in the United States and China, two large countries with a large number of pets, showed an obvious downward trend in 2020, and a significant rise in 2021. After that, the number of cats in the United States decreased year by year, while the number of cats in China increased year by year. In general, the number of cats in 2020 and 2021 showed the same trend as that in China and the United States, and then fluctuated in the following years. It can be predicted that the global cat industry will hit a trough in 2020, recover in 2021, and then develop steadily, with an overall increase compared with 2019.

Dogs: Also with reference to the above line chart, we can see that the number of dogs has not changed much either in various countries at home and abroad or in the world as a whole. Therefore, the global dog industry has shown a stable development trend in recent years.

Model building:

Since the development of the global pet food industry is affected by various countries and multiple factors, this paper considers the use of representative countries -- the United States, France and Germany to conduct research, and establishes the Spearman correlation analysis model to get the factors that are highly correlated with the demand of pet food market in various countries. Then, the gray forecasting model of the first question is used to obtain the predicted values of these factors in the next three years, as well as the predicted value of the proportion of pet food market size in these countries in the country. Due to the high volatility of the data and the small amount of data, it is difficult to use conventional multivariate fitting methods to establish the regression model. In this case, we consider using support vector machine regression to regression fit the pet market size of each country in the next three years, and calculate the global pet food market size according to the above proportion. To some extent, this scale can be regarded as an intuitive reflection of the global pet food market demand.

Spearman's correlation analysis model was established:

Spearman correlation coefficient is used to describe the degree of correlation between quantitative variables or sequencing variables. For samples with data capacity n, the correlation coefficient is obtained according to the following formula: $\rho$

$$\rho = \frac{\Sigma_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma_i (x_i - \bar{x})^2 \Sigma_i (y_i - \bar{y})^2}} \quad (1)$$

Where and represents the $x_i$ $y_i$ i th attribute value of the two sample datasets respectively, and is the sample average value. $\bar{x}$ $\bar{y}$

$\rho$ The change interval of is [-1, 1], with negative values representing negative correlation and positive values representing positive correlation. The closer the absolute value is to 1, the stronger the correlation is. $x_i$ $y_i$ $x_i$ $y_i$ In this topic, factors with relatively strong correlation with the pet market size of various countries are selected as independent variables to predict the pet food market size of various countries.

Support vector machine regression model establishment:

If there is a hyperplane (w,b), define the functional interval between the hyperplane and the sample point (,) $x_i$ $y_i$ $\hat{\gamma}_i = y_i(wx_i + b)$

The above formula is normalized, that is, the original w and b are replaced by and, and the normalized interval is defined as a geometric interval $\frac{w}{\|w\|}$ $\frac{b}{\|w\|}$

$$\gamma_i = y_i\left(\frac{w}{\|w\|}\cdot x_i + \frac{b}{\|w\|}\right)$$

Also, define the distance of a sample set to the classification hyperplane the geometric interval of the sample point closest to the classification hyperplane in this set

$$\delta = min\,\delta_i$$

When there are some specific points in the data set, the data set may be linearly indivisible. At this time, a relaxation variable >0 can be entered into each sample point (xi,yi), so that the function interval plus the relaxation variable is greater than or equal to 1, so that the constraint condition becomes$\xi_i y_i(w\cdot x_i + b) \geq 1 - \xi_i$

Construct the Lagrange function, introduce the Lagrange multiplier vector ≥0,i=1,2,3,4,5, define the Lagrange function L (w,b,) : $\alpha_i$ $\alpha$

$$min_a -\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j y_i y_j K(x_i, x_j) - \sum_{i=1}^{N}\alpha_i$$

$$s.t.\ \sum_{i=1}^{N}\alpha_j j_i = 0$$

The optimal solution is obtained$\alpha^* = (\alpha_1^*, x_2^*, \cdots, \alpha_N^*)^T$

(6) Select a positive component 0< to calculate$\alpha^*\alpha_j^* < C$

$$b^* = y_j - \sum_{i=1}^{\mathbb{N}}\alpha_i^* y_i K(x_i, x_j)$$

$$w^* x + b^* = 0$$

(7) Construct the decision function

$$f(x) = sgn(\sum_{j=1}^{N}\alpha_i^* y_i K(x, x_i) + b^*)$$

The size of the global pet food market is established:

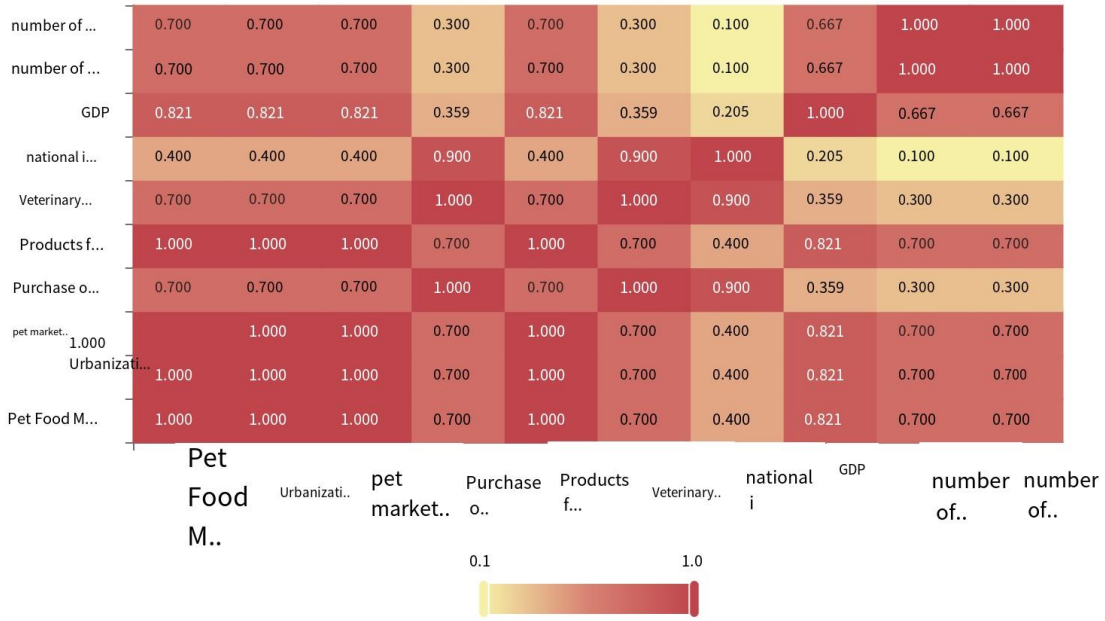
The predicted value of the pet food market size of t(t = 1,2,3) countries in the year k(k= 2,024,2525,2026) is denoted as, and the proportion of the pet food market size of each country in the global market predicted by the gray prediction model in the year k is comprehensively denoted as, thus the global pet food market size in the year k is as follows: $w_{kt}$ $p_k$ $s_k$

$$s_k = \frac{\sum_{t=0}^{3} w_{kt}}{p_k} \tag{2}$$

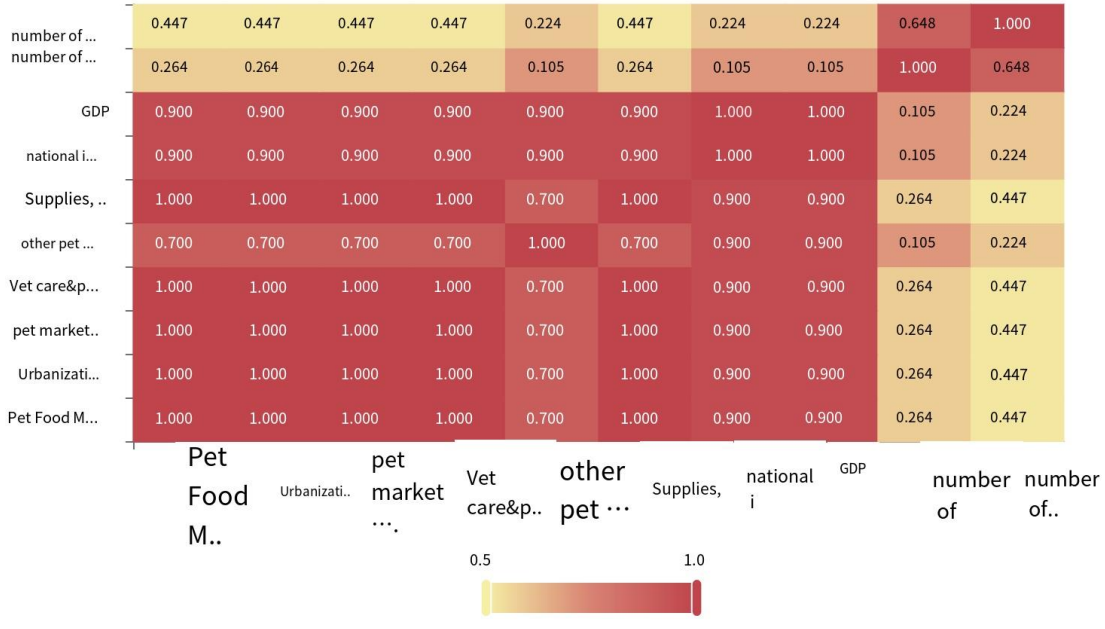
To solve the model:

Spearman correlation analysis model solving:

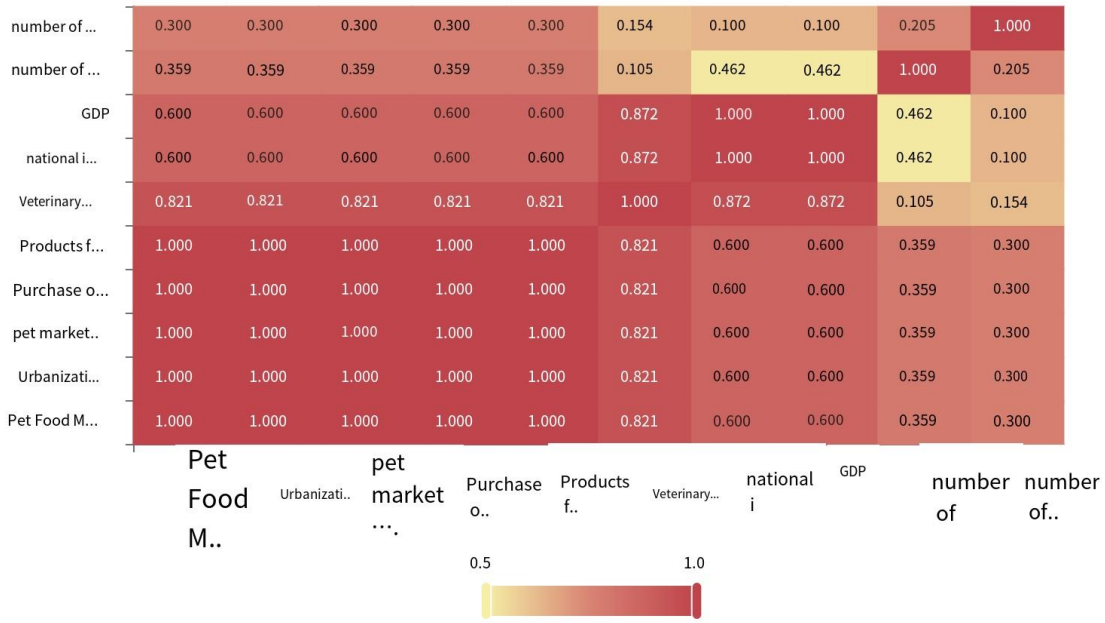
Taking France as an example, SPSS software was used to obtain the following heat map, in which the correlation between national income and other pet service and Pet Food Market was weak, so we selected several other indicators to further explore the changing trend of pet food market in various countries.




The United States:




Germany:




The final selected indicators are:

America:Pet Food Market,Urbanization Rate,pet market sales,Vet care&product sales,Supplies, live animals & OTC medicine,GDP,number of cats,number of dogs

Germany:Pet Food Market,Urbanization Rate,pet market sales,Products for pets,Veterinary and other services for pets,GDP,number of cats,number of dogs

France: Pet Food Market,Urbanization Rate,pet market sales,Products for pets,Veterinary and other services for pets,GDP,number of cats,number of dogs

Grey prediction model solution:

Follow the solution steps of the first question, substitute the data selected by each country, and get the forecast situation of each indicator in the next three years, see Appendix 2.

Support vector machine regression model is solved:

Through the above analysis, SPSS software is used to obtain the corresponding support vector machine regression model of each country, and the pet food market size of each country in the next three years is predicted according to the data obtained by the gray prediction model.

Model parameters:

| Parameter name | America | Germany | France |
|---|---|---|---|
| Training time | 0.008 s | 0.008 s | 0.008 s |
| Data segmentation | 0.8 | 0.8 | 0.8 |
| Data shuffling | is | is | is |
| Cross validation | 3 | 3 | 3 |
| Penalty factor | 1 | 1 | 1 |
| Kernel function | linear | linear | linear |
| Kernel coefficient | scale | scale | scale |
| Kernel constant | 0 | 0 | 0 |
| Highest degree of kernel function | 3 | 3 | 3 |
| Error convergence condition | 0.001 | 0.001 | 0.001 |

| | | | |
|---|---|---|---|
| Maximum number of iterations | 1000 | 1000 | 1000 |

Predicted results:

| Years | America | Germany | France |
|---|---|---|---|
| 2024 | 773.416982 | 63.69884522 | 60.3918744 |
| 2025 | 868.1027354 | 66.8604091 | 63.65835787 |
| 2026 | 964.6428793 | 71.09824122 | 66.89097114 |

Global pet food market demand solution:

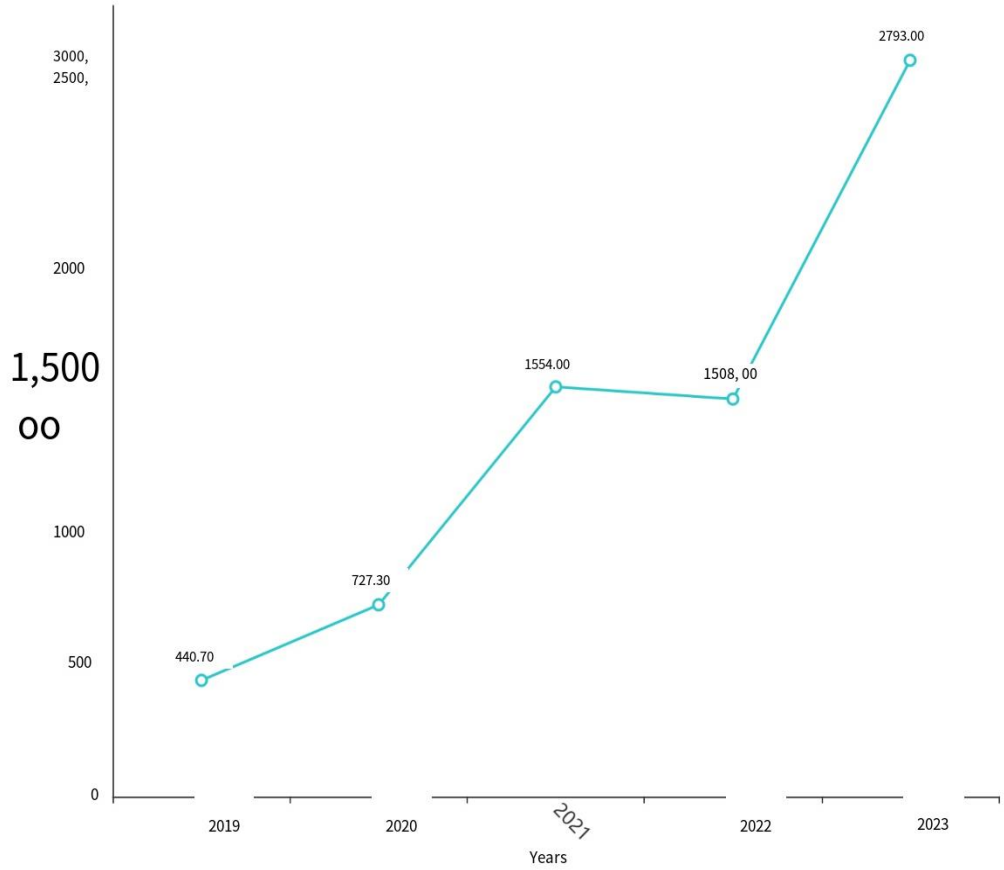
By substituting the data obtained from the support vector machine model into the established model, the global pet food market size can be obtained as follows. It can be approximated that the global pet food market size is the global pet food market demand.

| Years | Global pet food market size (million dollars) |
|---|---|
| 2024 | 773.416982 |
| 2025 | 868.1027354 |
| 2026 | 964.6428793 |

## 4.4 Model establishment and solution of Problem 3 (Analysis and Prospect of Pet Food Development in China based on Production and export, Level 1 title)

### 4.4.1 Development of Pet Food Industry in China (Level 2 Title)

#### IV.4.1.1 Analysis of Supply and Demand (Level 3 Title)



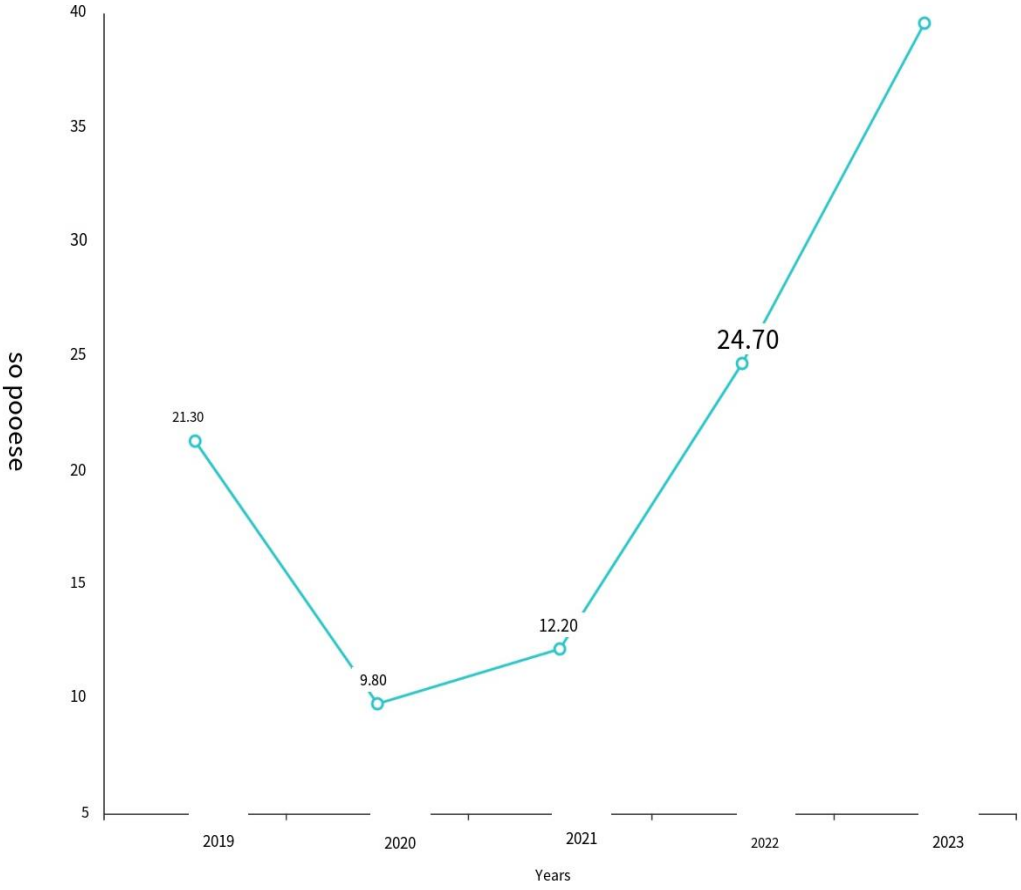

**Graph 5:1 Trend of Chinese Pet Food Production4-2**

The trend of pet food production in China from 2019 to 2023 is shown in graph (Figure). According to the trend change of production in five years, it is analyzed that there is a certain degree of fluctuation in pet food production in China, but the average growth rate reaches 65.48%, showing a fast rising trend as a whole. It can be seen that the development prospect of China's pet food industry is bright.

Among them, the growth rate in 2020-2021 and 2022-2023 is faster, reaching 113.67% and 85.21%, which is consistent with the growth trend of per capita disposable income in 2019-2023 we found, which may be related to the development of productivity level after the epidemic. From 2021 to 2022, the production of China's pet food industry showed a slight decline of 0.02%. According to the relevant documents and data, the epidemic situation in Shanghai and the whole country was repeated in early March 2022. The outbreak of the epidemic again led to the lack of materials in relevant regions, the shortage of personnel, the blockage of logistics, and the suspension of operation of some stores. From the supply side, pet products have become fragmented from production to transportation to marketing, and from the demand side,

the purchase of pet necessities has become a problem in the containment stage, so the analysis may be repeatedly related to the epidemic situation. At the same time, the pet dog market accounts for more than 40% of the domestic pet market, and the pet dog market will further shrink in 2021-2022. It is reasonable to believe that this may have an impact on the demand side of China's pet food industry, and then affect the supply side production.

#### IV.4.1.2 Analysis of Overseas influence and Competitiveness (three-level title)




**Graph V 2Trend of Chinese Pet Food Exports4-3**

As shown in the graph, China's pet food exports decreased in 2019-2020 and increased year by year in 2020-2023. It can be seen that excluding the impact of COVID-19 in 2019-2020, China's pet food exports continue to increase, and the relative competitiveness of China's pet food industry is constantly enhanced compared with other enterprises in the same industry from a global perspective. And its influence continues to expand. This growth momentum may be closely related to the development of China's science and technology, economy and transportation strength. The following model will quantify the above factors and specifically analyze the relationship between the world pet food market demand and China's development and export volume.

### 4.4.2 Prospect model of China's pet Food Production and Export in the future (Secondary title)

#### IV.4.2.1 Data Processing (Level 3 Heading)

For Pet food industry market size, Per Capita Disposable Income of Residents, Urbanization Rate, total import and export volume, Number of Valid Patent Authorizations in the Pet Industry, Transaction Scale of the Online Retail Industry Six data, we carried out visualization processing, get the following results.




**Graph Five 3 Possible influencing factors of pet food production value and export value in China4-4**

It can be seen that with the passing of time, the quantitative data in all aspects generally show an increasing trend, indicating that China's comprehensive national strength is improving year by year.

### IV.4.2.2 The establishment of the model (three levels of headings)

**Eigenvalue selection (Level 4)**

By default, a quantitative indicator () of global pet food market demand is selected, and four quantitative indicators of China's development with high correlation are selected to form an independent variable with dimension 5, which is used to represent the dependent variable China's pet food export value and the dependent variable China's pet food production value.

$$G_m \{C_1, C_2, C_3, C_4\} \; X^{(i)} \; (i = 2019, 2020, 2021, 2022, 2023) \; Y_e^{(i)} \; (i = 2019, 2020, 2021, 2022, 2023)$$

$$Y_p^{(i)} \; (i = 2019, 2020, 2021, 2022, 2023)$$

**SVR regression model**

After comprehensive analysis of the fluctuation trend of pet food production in China and the data quantified based on the trend of global pet food market demand and the development of China, it is impossible to determine the linear relationship between them. In order to ensure that the nonlinear relationship between various features can be taken into account in the process of model establishment to achieve a high precision modeling effect, SVR was selected to carry out the regression fitting relationship between each feature data (independent variable) and the amount of pet food production in China, and linear was selected as the kernel function, that is, the data was mapped to the high-dimensional data feature space by nonlinear mapping, so that the independent variables and dependent variables in the high-dimensional data feature space had good linear regression characteristics. After fitting the feature space and then returning to

the original space, the high-dimensional linear relationship formula was finally obtained, and the production amount predicted by the SVR model was obtained by substituting the gray prediction results. $X_{pre}^{(i)}(i=2024,2025,2026)$ $Y_{p_\mathrm{SVR}}^{(i)}$

**Three-fold cross-validation**

Due to the small sample size, we choose to use data shuffling, add a penalty factor of 1 to the model iteration process, and use three-fold verification to improve the reliability of the evaluation.

Step 1 Folding part: Divide the original data set into three parts with the same size as possible. In this case, five sets of characteristic values are obtained by combining the quantitative indicators of each year with the production amount. Then the data of 2019 and 2020 can be taken as a set, the data of 2021 and 2022 as a set, and the array of 2023 as a separate set. $\{(X^{(2019)},Y_p^{(2019)}),(X^{(2020)},Y_p^{(2020)}),(X^{(2021)},Y_p^{(2021)}),(X^{(2022)},Y_p^{(2022)}),(X^{(2023)},Y_p^{(2023)})\}$

Step 2 Model training and verification : Divided into three iterations, first select two parts as the training set and the rest as the verifier, use the verification set to evaluate the model performance and feedback for parameter changes. In each subsequent iteration, rotate the partitioning method, change the selected part of the verification set, and repeatedly conduct model training and evaluation until every subset has a chance to become the verification set, and complete the last training and verification process.

Step 3 Performance evaluation: Calculate and take the average value of the three validation results as the final performance index of the model. This model uses the mean square error MSE, $R^2$ evaluation index.

**Partial least squares regression model (PLSR)**

It is found that the export value of pet food in China in the past five years, except for the trend of decline between 2019-2020, the export value and the correlation between 2021-2023 are linear, and may present multicollinearity. $Y_e^{(i)}$ $X^{(i)}$ $(i=2020,2021,2022,2023)$ Therefore, the partial least square (PLSR) regression model suitable for multi-input, small sample size and possibly multicollinearity is chosen here for modeling. The modeling principle is the same.

**Grey prediction Model (Level 4 title)**

For example, in the steps of problem 1 and Problem 2, the grey prediction model is used to complete the prediction of the independent variables of the 5 regression models after screening and the dependent variables of the two regression models of production volume and export volume. $X_{pre}^{(i)}(i=2024,2025,2026)$ $Y_{e_gray}^{(i)}$ $Y_{p_gray}^{(i)}(i=2024,2025,2026)$

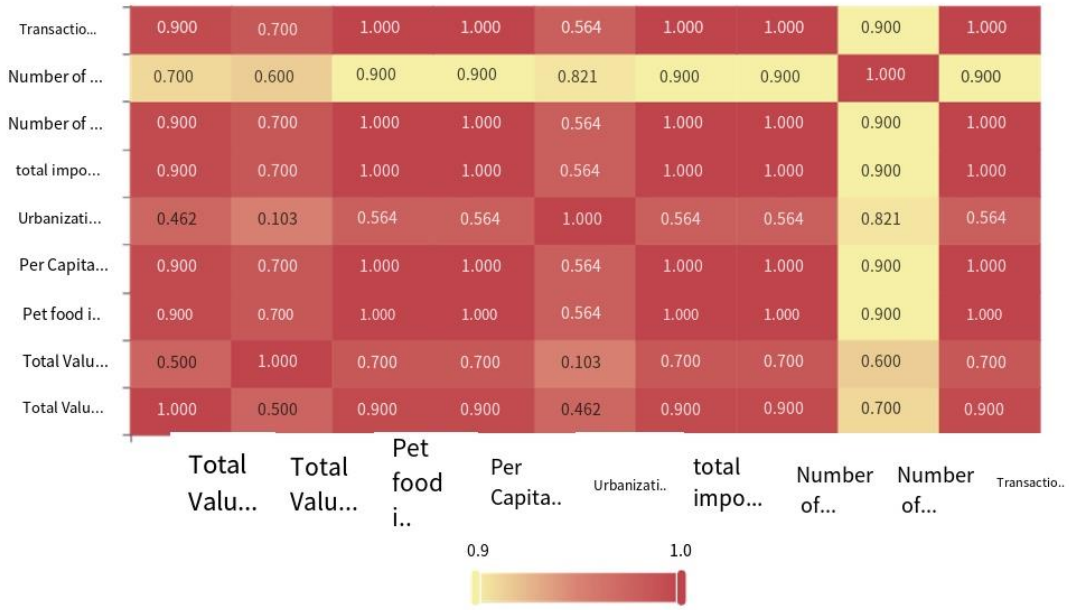
**Prediction regression fitting**

Plug in the linear polynomial obtained by the PLSR to get the export value predicted by

the partial least squares regression model. $X_{pre}^{(i)}(i=2024,2025,2026)$ $Y_{e_pls}^{(i)}$

#### IV.4.2.3 Solving the model (three-level heading)

(1) Correlation solution results




According to Spearman correlation analysis, five characteristic variables with high correlation were screened, namely, Pet food industry market size, Per Capita Disposable Income of Residents, total import and export volume, Number of Valid Patent Authorizations in the Pet Industry, Transaction Scale of the Online Retail Industry Five variables are used as independent variables for the SVR and PLSR models.

(2) Grey prediction solution results

The gray prediction of independent variables all passed the grade-ratio test, and the detection results of the prediction results were all good. The data after the gray prediction was obtained are shown in the following table

| Years | 2024 | 2025 | 2026 |
|---|---|---|---|
| Pet food industry market size | 1464.857 | 1595.687 | 1738.201 |
| Per Capita Disposable Income of Residents | 41897.143 | 44641.483 | 47565.582 |
| total import and export volume | 468176.767 | 506616.443 | 548212.209 |
| Number of Valid Patent Authorizations in the Pet Industry | 10598 | 12478 | 14545 |
| Transaction Scale of the Online Retail Industry | 166004.297 | 180308.05 | 195844.286 |

The grey effect of dependent variable export value is shown in the table, with an average relative error of 13.231%.

| Indexed Items | Original value | Predicted value | Residual | Relative error (%) |
|---|---|---|---|---|
| 2019 | 21.3 | 21.3 | 0 | 0 |
| 2020 | 9.8 | 6.529 | 3.271 | 33.381 |
| 2021 | 12.2 | 15.225 | -3.025 | 24.795 |
| 2022 | 24.7 | 25.547 | -0.847 | 3.428 |
| 2023 | 39.6 | 37.797 | 1.803 | 4.552 |

The grey effect of dependent variable export value is shown in the table, with an average relative error of 10.495%

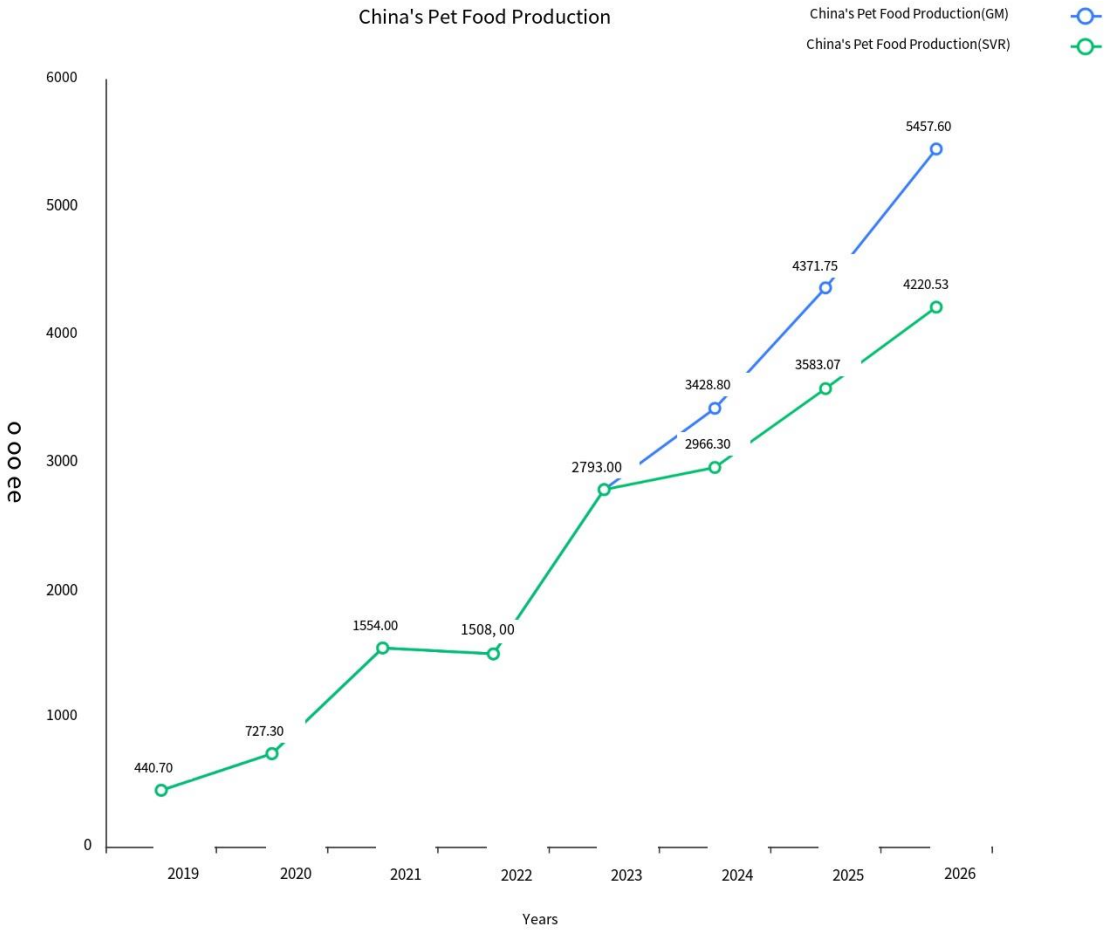
| Index entry | Original value | Predicted value | Residual | Relative error (%) |
|---|---|---|---|---|
| 2019 | 440.7 | 440.7 | 0 | 0 |
| 2020 | 727.3 | 745.166 | -17.866 | 2.457 |
| 2021 | 1554 | 1281.391 | 272.609 | 17.542 |
| 2022 | 1508 | 1898.884 | -390.884 | 25.921 |
| 2023 | 2793 | 2609.96 | 183.04 | 6.554 |

(3) SVR forecast production in the next three years change trend solution results

| Evaluation index | Evaluation results |
|---|---|
| $R^2$ | 0.9506751950629758 |
| MAPE | 15.646846073577029 |

| $Y_{p_SVR}^{(i)}$ (i=2019-2023) | Total Value of China's Pet Food Production | Pet food industry market size | Per Capita Disposable Income of Residents | total import and export volume | Number of Valid Patent Authorizations in the Pet Industry | Transaction Scale of the Online Retail Industry |
|---|---|---|---|---|---|---|
| 585.3829825039247 | 440.7 | 962 | 30733 | 315627.3 | 2271 | 103200 |
| 871.3583505938986 | 727.3 | 1028 | 32189 | 322215.2 | 4118 | 119213 |
| 1273.3454045470098 | 1554 | 1151 | 35128 | 387391.8 | 6311 | 131419 |
| 1394.8431856970528 | 1508 | 1237 | 36883 | 416727.8 | 7888 | 137853 |
| 2614.3749329040984 | 2793 | 1340 | 39218 | 417510.1 | 8458 | 154264 |

Compared with gray prediction, SVR has smaller relative error to the fitting results of the existing five-year dependent variable data, so SVR model is more suitable.




According to the SVR forecast results, the change trend of China's pet food production in

the next three years is increasing. The specific value is shown in the figure, which is 296.63 billion yuan in 2024, 358.3 billion yuan in 2025 and 4,220.53 yuan in 2026.

(4) Solution result of PLSR forecast export volume

Set three principal components to be extracted, and the regression equation is as follows:

Total Value of China's Pet Food Exports = 21.52+31.255 * Pet food industry market size(standardized)+26.449 * Per Capita Disposable Income of Residents(standardized)-0.396 * total import and export volume(standardized)-34.461 * Number of Valid Patent Authorizations in the Pet Industry(standardized)-15.409 * Transaction Scale of the Online Retail Industry(standardized)

| | Total Value of China's Pet Food Exports |
|---|---|
| Constant | 21.52 |
| Pet food industry market size | 31.255 |
| Per Capita Disposable Income of Residents | 26.449 |
| total import and export volume | -0.396 |
| Number of Valid Patent Authorizations in the Pet Industry | -34.461 |
| Transaction Scale of the Online Retail Industry | -15.409 |
| $R^2$ | 0.941 |

After replacing the data of the previous five years, the result test and analysis obtained a value of 0.941 close to 1, which is better than the gray prediction model. $R^2$

After the independent variables of gray prediction are standardized by the improved version of Z-score similar to the first question, the above formula can be substituted, and the change trend of China's pet food export volume in the next three years can be obtained.

Model building

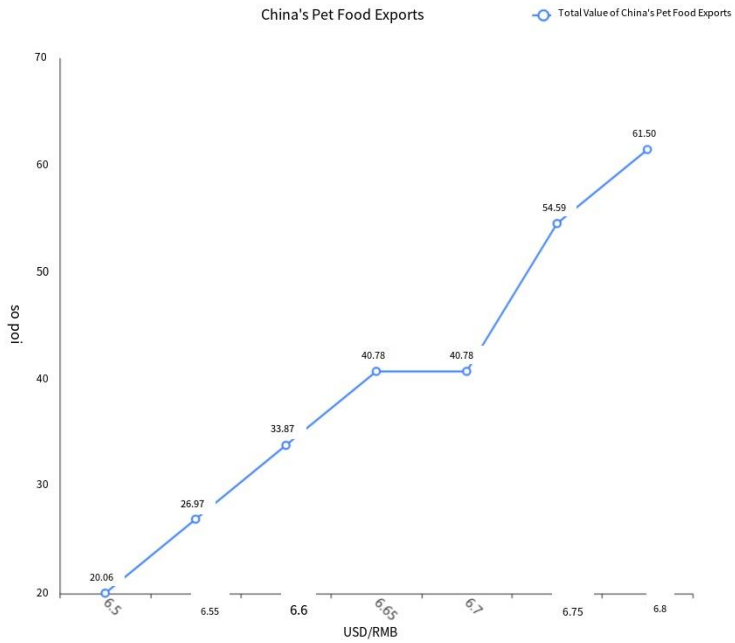
In order to quantify the impact of foreign economic policies on China's pet food industry, this paper follows the partial least squares model constructed by Question 3 on the export volume of pet food in China, and only adds the exchange rate of USD to RMB to obtain a new partial least squares model. Then, other factors are kept unchanged in 2023 and only the exchange rate is changed. In order to obtain the quantitative impact of China's pet food export volume under different exchange rates of US dollar to RMB, and it is approximately regarded as the impact on China's pet food market.

Model solving

Using SPSS software and relevant codes, we obtained the impact of exchange rate on the export volume of pet food in China as follows:

| USD/RMB | Total Value of China's Pet Food Exports |
|---|---|
| 6.5 | 20.06095 |
| 6.55 | 26.966905 |
| 6.6 | 33.872860 |
| 6.65 | 40.778815 |
| 6.7 | 40.778815 |

| | |
|---|---|
| 6.75 | 54.590725 |
| 6.8 | 61.49668 |




ENTERPRISES

First, companies should be keenly aware of policy changes and establish a complete evaluation system to assess the impact of different policy changes on pet food price changes, market demand, etc., in order to respond appropriately when the new policy has just been introduced.

Secondly, in the face of the implementation of the new policy, enterprises need to timely use financial analysis and market research and other means to conduct a comprehensive impact assessment again, accurately quantify the size of the pros and cons of policy changes on business operations, based on the evaluation results, to develop a more scientific and reasonable pricing strategy.

In addition, in order to deal with the market changes caused by policies more effectively, enterprises should establish a set of flexible market feedback mechanism. By monitoring the impact of price changes on market demand in real time, pricing strategies should be adjusted accordingly. At the same time, price elasticity analysis to understand the sensitivity of different markets to price changes is helpful for enterprises to develop more accurate and differentiated pricing strategies.

Finally, the overseas pet food industry is developing rapidly, and enterprises should actively explore new markets. Companies should make full use of free trade agreements and other international trade policies to reduce or avoid tariff barriers and lower entry costs. At the same time, they should conduct a comprehensive assessment of potential risks in new markets and formulate corresponding risk management strategies to reduce their dependence on the single market and ensure robust business development.

The State

The state should actively adjust the domestic pet food industry structure, optimize the export

product structure and market layout, and at the same time actively seek economic and trade cooperation with other countries, promote industrial upgrading and transformation, and improve the self-regulation ability of the pet food industry in the face of new economic policies.

At the same time, in the face of the impact of new economic policies in other countries, fiscal and financial policies can be adopted to support the pet food enterprises that have been greatly affected, reduce the burden of enterprises and promote their development.

In the face of unreasonable and unfair economic policies of other countries, the country can negotiate, and when necessary, it can take countermeasures, including but not limited to taking anti-dumping and countervailing measures, and when necessary, it can impose tariffs on imported goods from other countries, in order to maintain national interests and economic and trade order.

# 5 Model Evaluation and Further Discussion

## 5.1 Strengths

### 5.1.1 Diverse and appropriate application of models

This paper comprehensively considers a variety of models, and chooses a more suitable model for each question. Finally, Pearson correlation coefficient analysis model, spearman correlation coefficient analysis model, gray prediction model, partial least squares model, support vector machine model and improved z-score standardized model are adopted. And for some data prediction, this paper uses a variety of models, and their output results are compared and analyzed, in order to find the best results. It not only realizes the diversification of models, but also improves the accuracy of prediction.

### 5.1.2 Flexible definitions of standardization

In this paper, z-score standardized model is innovatively modified, and the modified model is combined with partial least squares model and gray prediction model to achieve an innovative method for the prediction of data affected by multiple factors.

## 5.2 Weaknesses

### 5.2.1 More factors considered

In the second question to forecast the global demand for pet food, we only consider the relevant changes of the three countries with the largest demand for pet food, the United States, France and Germany. In order to better forecast the global demand for pet food, we can also consider China, the United Kingdom and other countries with the same demand for pet food, so as to obtain more accurate forecast results.

## 5.3 Further Discussion

### 5.3.1 The application of entropy method

In the analysis of the development of a variable, we rely on a variety of indicators to carry out analysis. For example, in the first question of analyzing the development of China's pet industry by pet type, we analyzed the number of pets and the size of the pet market. In order to realize the comprehensive evaluation of multiple indicators, we can adopt the entropy method. Entropy method refers to the mathematical method used to judge the degree of dispersion of an index. The greater the degree of dispersion of an index, the greater the degree of randomness and disorder, and the greater the impact on the comprehensive evaluation. We can calculate the entropy value of each index to obtain their weight, to carry out comprehensive evaluation.

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# Appendices

**Appendix 1**

Introduce: Solution results of grey prediction model for problem one

| Years | 2024 | 2025 | 2026 |
|---|---|---|---|
| Number of Dogs | 5125 | 5081 | 5038 |
| Number of Cats | 9486 | 11451 | 13581 |
| Scale of the Dog Consumption Market | 1642.568 | 1756.348 | 1878.01 |
| Scale of the Cat Consumption Market | 1515.528 | 1717.839 | 1947.156 |
| Urbanization Rate | 0.657 | 0.658 | 0.66 |
| Proportion of the Single Population Aged 15 and Above in the Total Population | 0.292 | 0.297 | 0.303 |
| Cumulative Number of Current National-level Development Policies and Plans for the Pet Industry | 39.774 | 43.17 | 46.718 |
| Per Capita Disposable Income of Residents | 41897.143 | 44641.483 | 47565.582 |
| Proportion of the Population Aged 65 and Above | 0.162 | 0.169 | 0.177 |
| Transaction Scale of the Online Retail Industry | 166004.297 | 180308.05 | 195844.286 |
| Number of Valid Patent Authorizations in the Pet Industry | 10598 | 12478 | 14545 |
| Number of Financing Events in the Pet Industry | 34 | 29 | 24 |
| Number of Newly Established Enterprises with Business Scope Containing "Pet" | 240 | 312 | 397 |
| Number of Licensed Veterinarians | 18 | 19 | 20 |

**Appendix 2**

Introduce: Solution results of grey prediction model for problem otwo

America:

| Years | 2024 | 2025 | 2026 |
|---|---|---|---|
| Pet Food Market(US) | 773.416982 | 868.1027354 | 964.6428793 |
| Urbanization Rate | 83.1 | 83.201 | 83.301 |
| pet market sales | 166.935 | 186.406 | 208.146 |
| Vet care&product sales | 40.878 | 43.559 | 46.416 |
| Supplies, live animals & OTC medicine | 37.104 | 41.146 | 45.627 |
| GDP | 73632.825 | 75526.282 | 77468.428 |

| | | | |
|---|---|---|---|
| number of cats | 7815 | 7874 | 7933 |
| number of dogs | 8267 | 8127 | 7994 |

Germany:

| Years | 2024 | 2025 | 2026 |
|---|---|---|---|
| Pet Food Market(Germany) | 63.69884522 | 66.8604091 | 71.09824122 |
| Urbanization Rate | 77.9 | 78.001 | 78.101 |
| pet market sales | 64.148 | 37.602 | 71.242 |
| Products for pets | 141.446 | 155.126 | 170.13 |
| Veterinary and other services for pets | 163.911 | 184.394 | 207.436 |
| GDP | 55452.874 | 56346.973 | 57255.487 |
| number of cats | 1546 | 1531 | 1517 |
| number of dogs | 1045 | 1042 | 1039 |

France:

| Years | 2024 | 2025 | 2026 |
|---|---|---|---|
| Pet Food Market(France) | 60.3918744 | 63.65835787 | 66.89097114 |
| Urbanization Rate | 81.601 | 81.803 | 82.005 |
| pet market sales | 125.47 | 130.321 | 135.36 |
| Products for pets | 117.275 | 120.695 | 124.215 |
| Veterinary and other services for pets | 47275.731 | 48767.361 | 50306.054 |
| GDP | 1666.32 | 1721.359 | 1778.215 |
| number of cats | 1666 | 1721 | 1778 |
| number of dogs | 1007 | 1096 | 1194 |